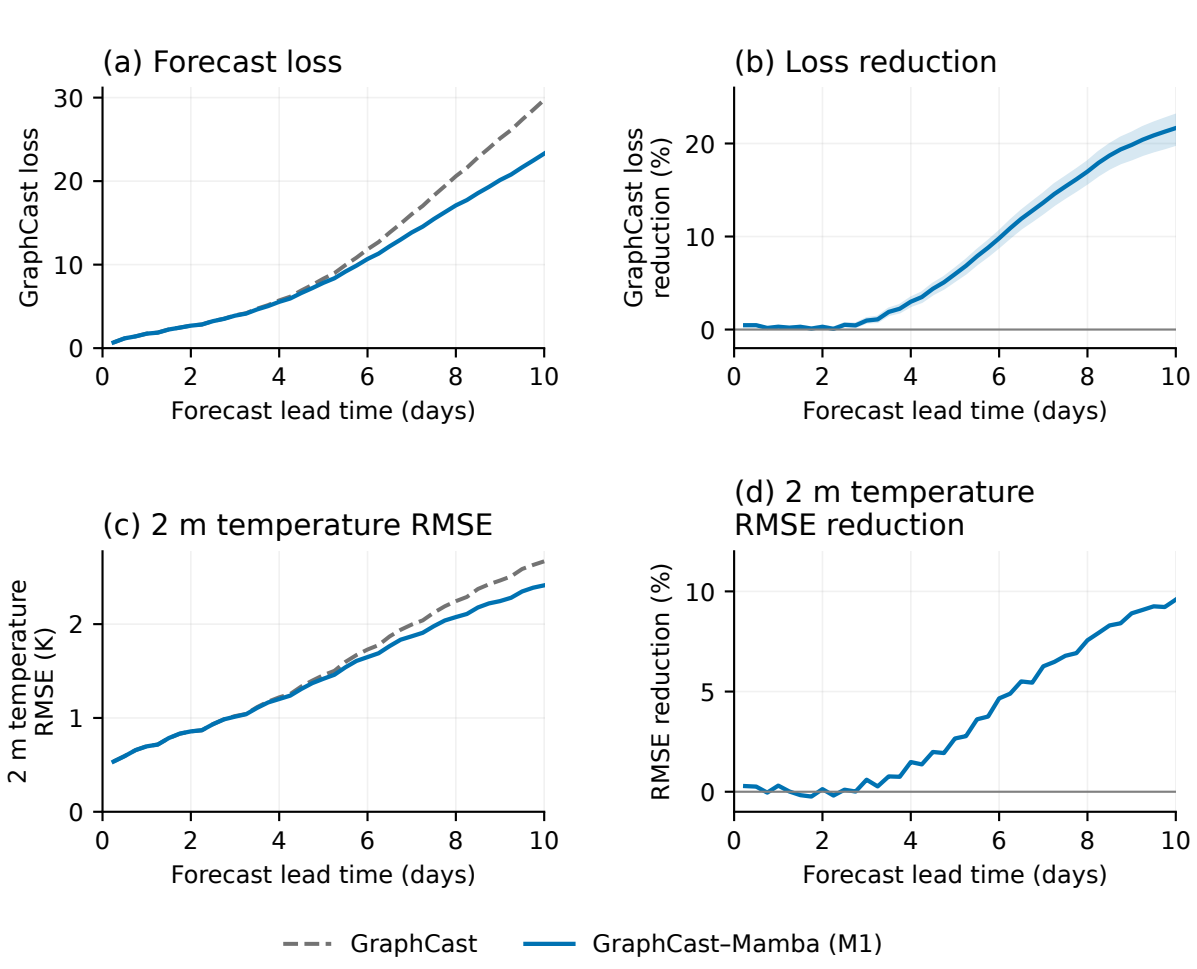
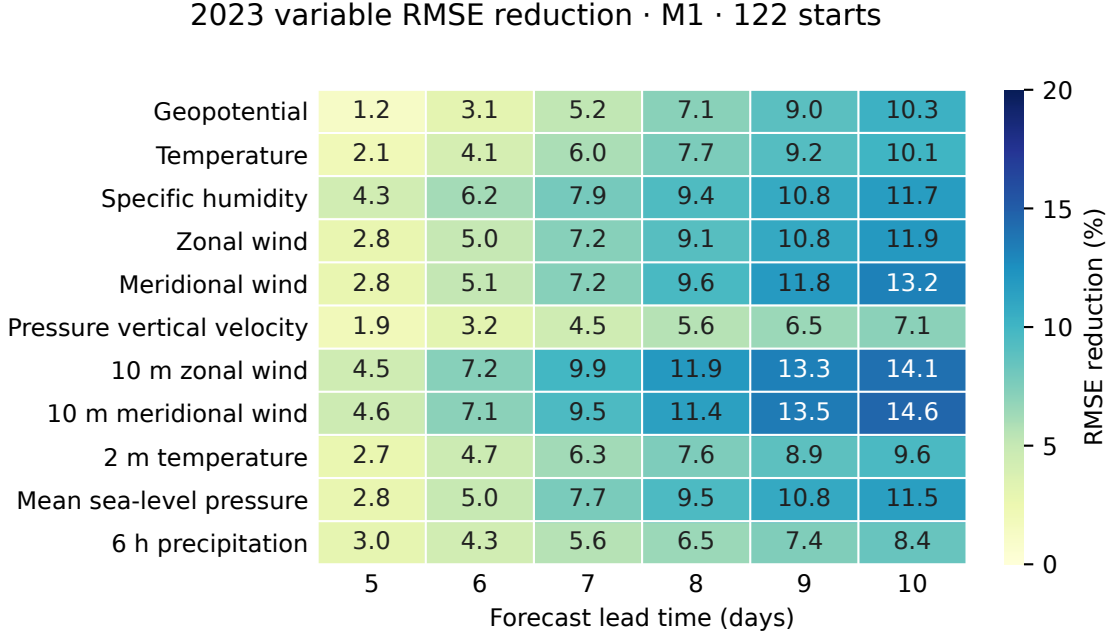
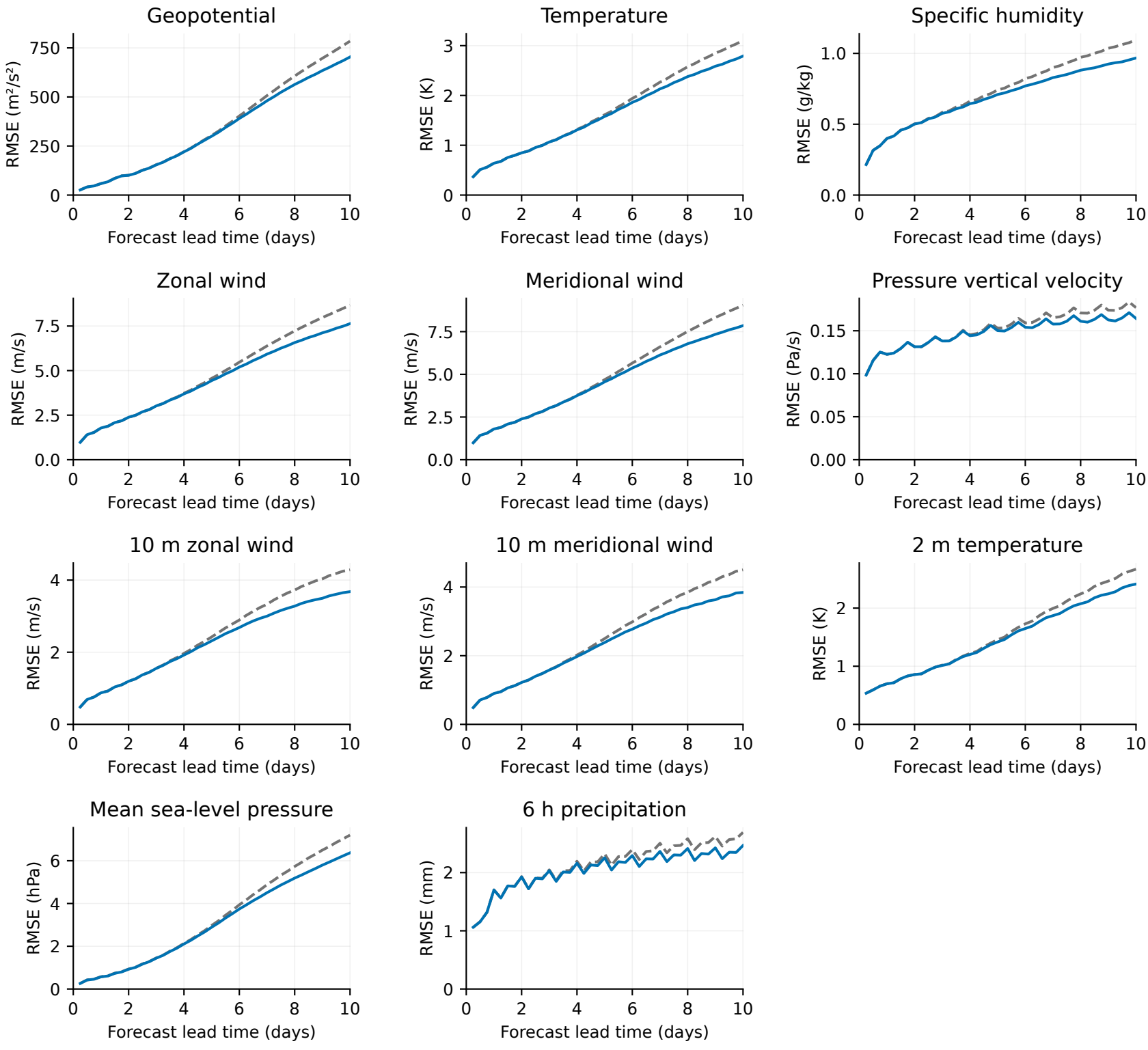


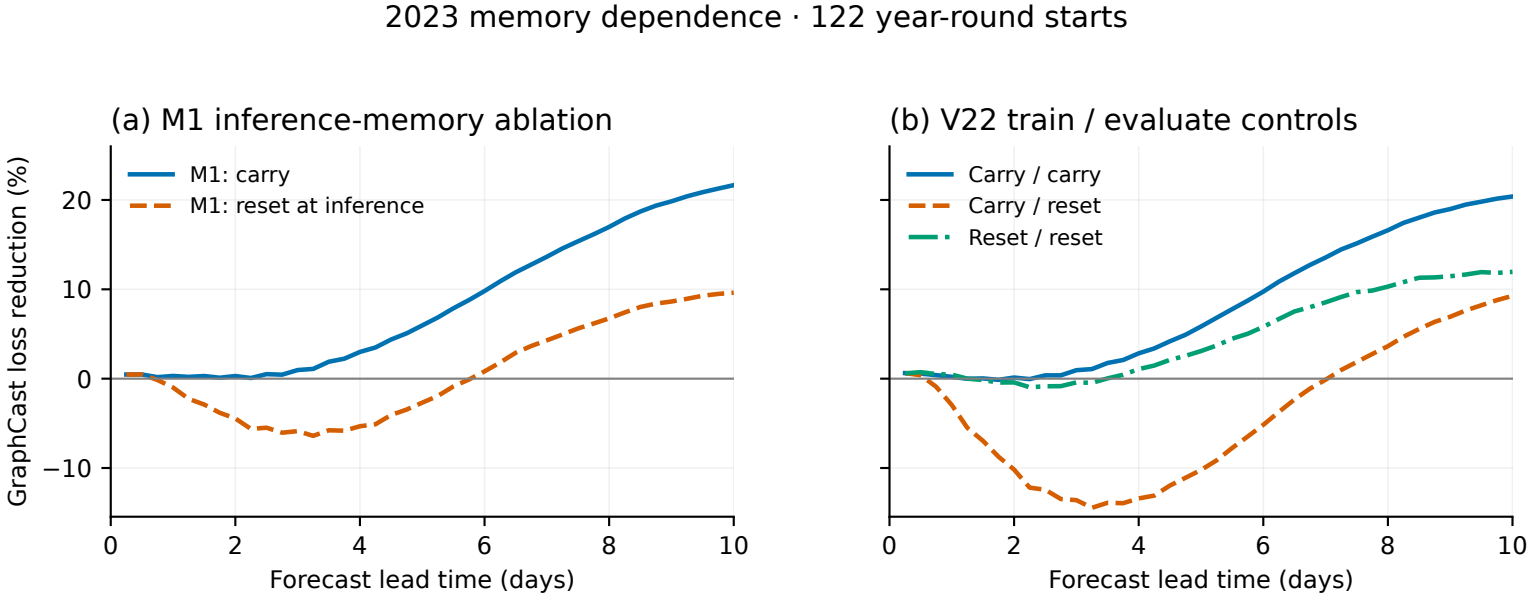
Exact loss reduction: rollout 14.42% · day 10 21.66%

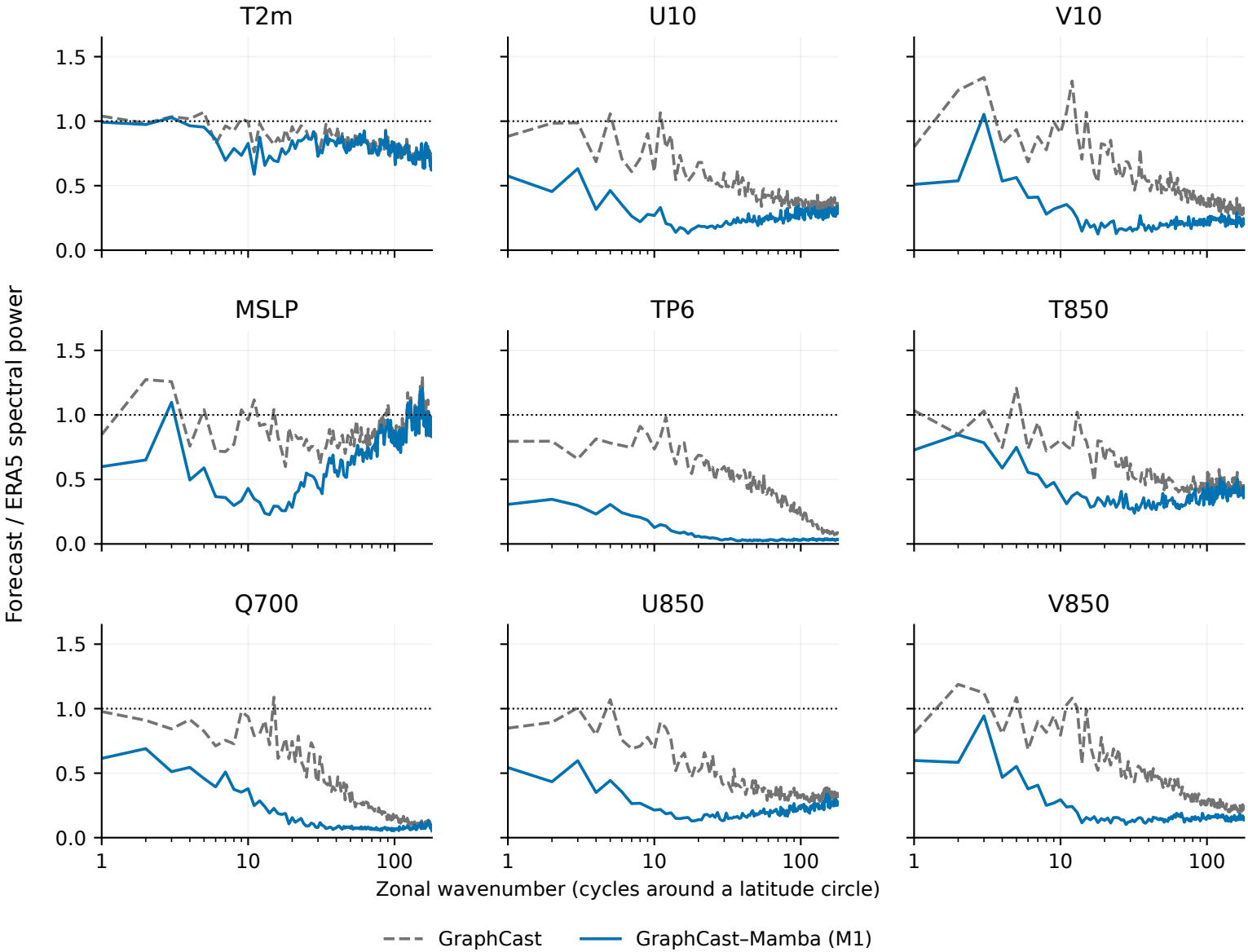
Shading: pointwise 95% loss intervals; numerical-repeatability warning retained.



Physical MSE pooled before square root; upper-air groups use equal pressure levels.


2023 physical-unit RMSE · M1 · 122 year-round starts
Geopotential
RMSE (m²/s²)
Temperature
RMSE (K)
Specific humidity
RMSE (g/kg)
Zonal wind
RMSE (m/s)
Meridional wind
RMSE (m/s)
Pressure vertical velocity
RMSE (Pa/s)
10 m zonal wind
RMSE (m/s)
10 m meridional wind
RMSE (m/s)
2 m temperature
RMSE (K)
Mean sea-level pressure
RMSE (hPa)
6 h precipitation
RMSE (mm)
Forecast lead time (days)
GraphCast
GraphCast–Mamba (M1)


2023 memory dependence · 122 year-round starts
(a) M1 inference-memory ablation
M1: carry
M1: reset at inference
GraphCast loss reduction (%)
20
10
0
−10
0
2
4
6
8
10
Forecast lead time (days)
(b) V22 train / evaluate controls
Carry / carry
Carry / reset
Reset / reset
0
2
4
6
8
10
Forecast lead time (days)
Exact original GraphCast loss throughout; common canonical baseline.
V22 controls are separate training runs; they are not matched capacity controls for M1.


2023 day-10 spectral power · eight-start pilot
Midlatitude zonal anomalies (30–60° N/S); unsmoothed positive modes
T2m
U10
V10
MSLP
TP6
T850
Q700
U850
V850
Forecast / ERA5 spectral power
0.0
0.5
1.0
1.5
1
10
100
Zonal wavenumber (cycles around a latitude circle)
GraphCast
GraphCast–Mamba (M1)
Dotted line: ERA5 power. Separate pilot sample; no uncertainty or preservation claim.